

Customer Behaviour Analysis for Alfido-Tech

This project focuses on analyzing customer purchasing behavior using data-driven techniques to uncover key revenue drivers, spending patterns, and retention trends. It involves exploratory data analysis (EDA), customer segmentation (RFM), churn analysis, and cohort-based retention modeling to generate actionable business insights. The goal is to enable strategic decision-making by identifying high-value customers, optimizing product performance, and improving customer retention through targeted strategies.

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
sns.set(style="whitegrid")

import warnings
warnings.filterwarnings('ignore')
```

```
In [2]: df = pd.read_csv("/content/ecommerce_customer_data_large.csv")
df
```

Out[2]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31
...	...	...	...	...	...	...	...	...
249995	33807	2023-01-24 12:32:18	Home	436	1	3664	Cash	63
249996	20455	2021-06-04 05:45:25	Electronics	233	1	4374	Credit Card	66
249997	28055	2022-11-10 17:11:57	Electronics	441	5	5296	Cash	63
249998	15023	2021-06-27 14:42:12	Electronics	44	2	2517	Cash	64
249999	4148	2020-09-07 05:12:19	Home	307	5	3634	Cash	32

250000 rows × 13 columns



In [3]: df.head()

Out[3]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	

In [4]:

```
print("Shape of dataset:", df.shape)
print("\nColumns:\n", df.columns)
```

Shape of dataset: (250000, 13)

Columns:

```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
      'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
      'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```

Data Overview

In [5]:

```
print(" Dataset Information:\n")
df.info()

print("\n Statistical Summary:\n")
display(df.describe())

print("\n Missing Values:\n")
print(df.isnull().sum())
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                           250000 non-null int64
1   Purchase Date                         250000 non-null object
2   Product Category                     250000 non-null object
3   Product Price                        250000 non-null int64
4   Quantity                             250000 non-null int64
5   Total Purchase Amount                 250000 non-null int64
6   Payment Method                       250000 non-null object
7   Customer Age                         250000 non-null int64
8   Returns                              202618 non-null float64
9   Customer Name                        250000 non-null object
10  Age                                  250000 non-null int64
11  Gender                              250000 non-null object
12  Churn                               250000 non-null int64
dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB
```

Statistical Summary:

	Customer ID	Product Price	Quantity	Total Purchase Amount	Customer Age	Returns
count	250000.000000	250000.000000	250000.000000	250000.000000	250000.000000	202618.000000
mean	25017.632092	254.742724	3.004936	2725.385196	43.798276	0.500000
std	14412.515718	141.738104	1.414737	1442.576095	15.364915	0.500000
min	1.000000	10.000000	1.000000	100.000000	18.000000	0.000000
25%	12590.000000	132.000000	2.000000	1476.000000	30.000000	0.000000
50%	25011.000000	255.000000	3.000000	2725.000000	44.000000	1.000000
75%	37441.250000	377.000000	4.000000	3975.000000	57.000000	1.000000
max	50000.000000	500.000000	5.000000	5350.000000	70.000000	1.000000

Missing Values:

Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	47382
Customer Name	0
Age	0
Gender	0
Churn	0

dtype: int64

Data Overview Insights :

The dataset contains ~250,000 records, indicating a large-scale customer transaction dataset suitable for analysis. There are 13 columns, including customer demographics, purchase details, and churn information. Most columns are complete, but the 'Returns' column contains missing values, which may require handling. The dataset includes a mix of: Numerical features (Price, Quantity, Age, Total Purchase) Categorical features (Product Category, Payment Method, Gender) Summary statistics show: Average purchase amount ≈ 2700 Average customer age ≈ 44 years No immediate anomalies in min/max values, indicating relatively clean data

DATA-CLEANING

```
In [6]: print(" ♦ Missing Values Before Cleaning:\n")
print(df.isnull().sum())

numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns

for col in numeric_cols:
    df[col].fillna(df[col].median(), inplace=True)

categorical_cols = df.select_dtypes(include=['object']).columns

for col in categorical_cols:
    df[col].fillna(df[col].mode()[0], inplace=True)

duplicates = df.duplicated().sum()
print("\n ♦ Duplicate Rows:", duplicates)

df.drop_duplicates(inplace=True)

print("\n ♦ Missing Values After Cleaning:\n")
print(df.isnull().sum())

print("\n ♦ Dataset Shape After Cleaning:", df.shape)
```

◆ Missing Values Before Cleaning:

```
Customer ID          0
Purchase Date        0
Product Category     0
Product Price        0
Quantity             0
Total Purchase Amount 0
Payment Method       0
Customer Age         0
Returns              47382
Customer Name        0
Age                  0
Gender               0
Churn                0
dtype: int64
```

◆ Duplicate Rows: 0

◆ Missing Values After Cleaning:

```
Customer ID          0
Purchase Date        0
Product Category     0
Product Price        0
Quantity             0
Total Purchase Amount 0
Payment Method       0
Customer Age         0
Returns              0
Customer Name        0
Age                  0
Gender               0
Churn                0
dtype: int64
```

◆ Dataset Shape After Cleaning: (250000, 13)

Data Cleaning Insights :

Missing values were identified primarily in the 'Returns' column, which were handled using median imputation for numerical stability. Categorical columns were cleaned using mode imputation, ensuring no data loss. Duplicate records were checked and removed to maintain data integrity. After cleaning: The dataset is fully complete (no null values) Data is consistent and ready for analysis This step ensures accurate insights and reliable modeling

Exploratory Data Analysis

```
In [7]: df_sample = df.sample(2000, random_state=42)
num_cols = ['Product Price', 'Quantity', 'Total Purchase Amount', 'Customer Age']

plt.figure(figsize=(12, 8))
```

```

for i, col in enumerate(num_cols, 1):
    plt.subplot(2, 2, i)
    plt.hist(df_sample[col], bins=30)
    plt.title(col)

plt.tight_layout()
plt.show()

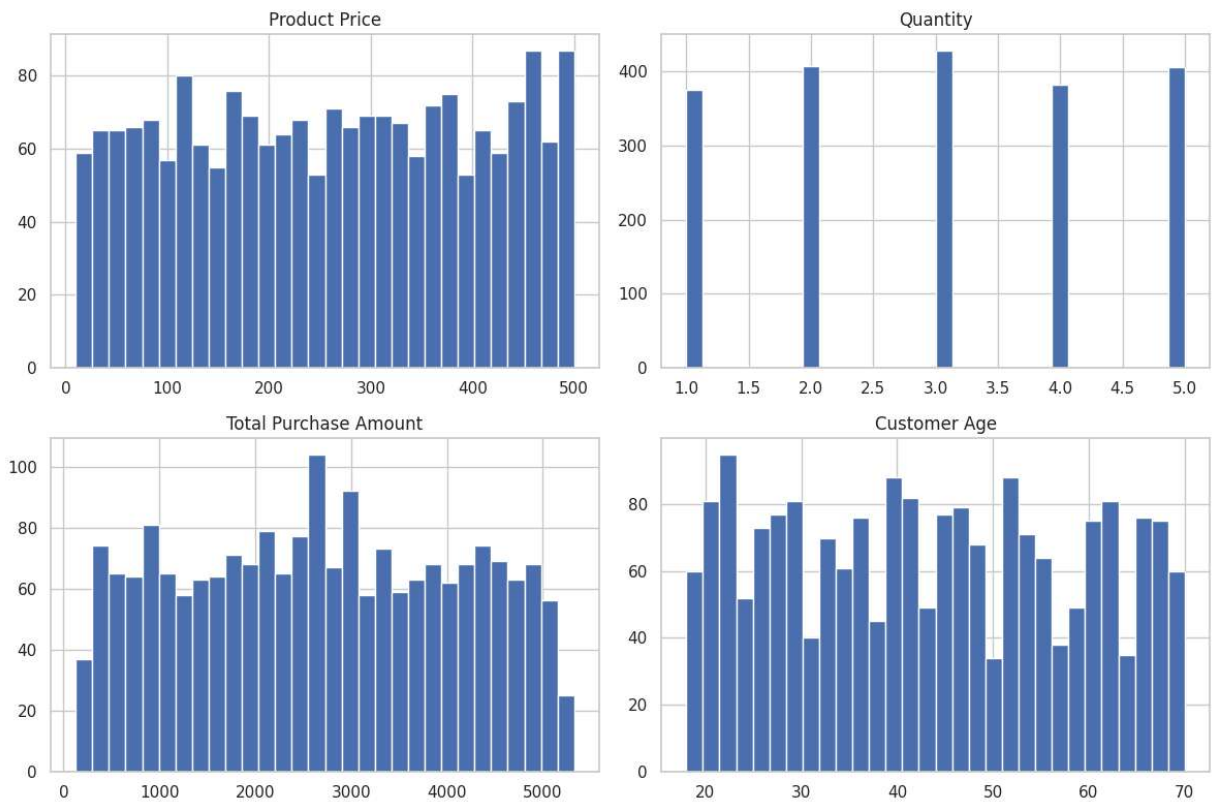
cat_cols = ['Product Category', 'Payment Method', 'Gender', 'Churn']

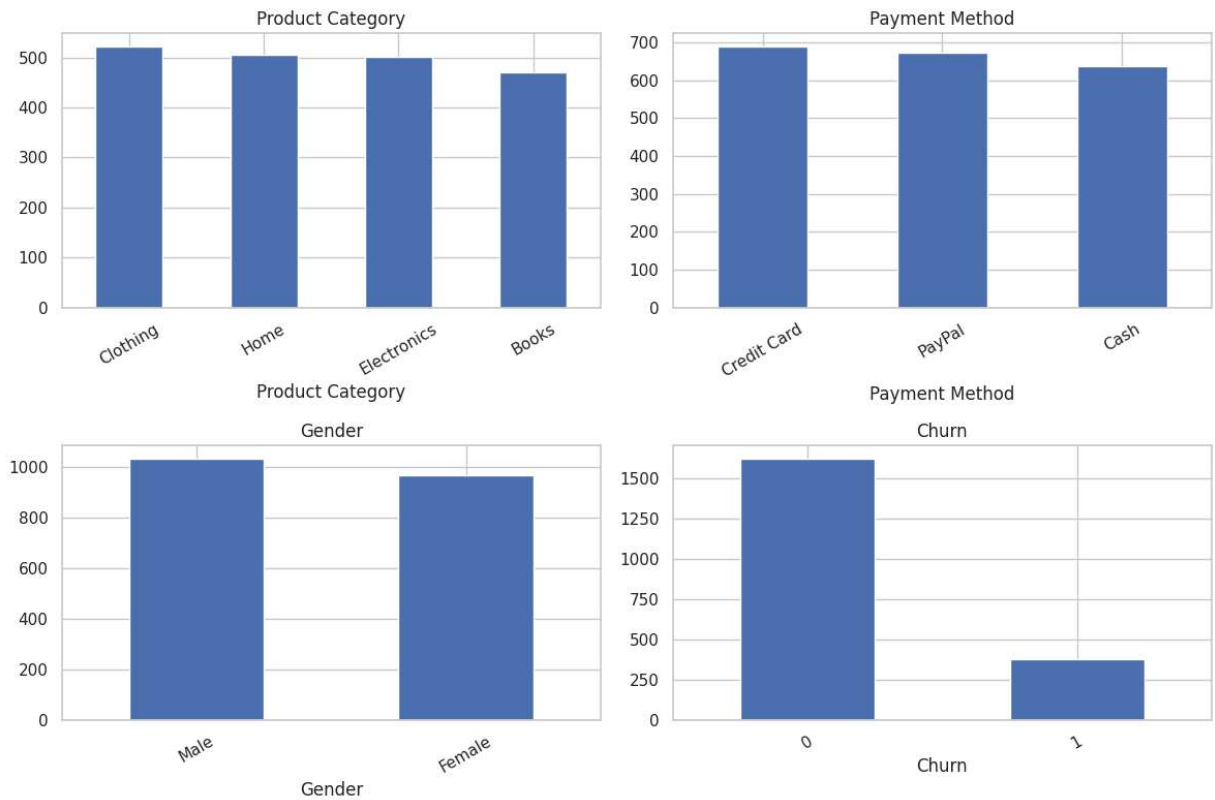
plt.figure(figsize=(12, 8))

for i, col in enumerate(cat_cols, 1):
    plt.subplot(2, 2, i)
    df_sample[col].value_counts().plot(kind='bar')
    plt.title(col)
    plt.xticks(rotation=30)

plt.tight_layout()
plt.show()

```





Numerical Features — Deep Insights :

Most numerical features exhibit near-normal distributions, indicating stable purchasing behavior across customers. Total Purchase Amount shows a strong central tendency (~2700), suggesting a consistent spending pattern among users. Quantity is concentrated between 1–5, indicating: Customers prefer low-volume, frequent purchases Business model is likely transaction-driven rather than bulk-driven Product Price has a wide spread → reflects a diverse product catalog (low to premium range) Customer Age is centered around 30–55, implying: Core customers belong to working-class / economically active segment Strong purchasing power group

👉 Business Insight:

The platform primarily serves mid-age, consistent-spending customers with moderate purchase frequency.

Categorical Features — Behavioral Insights :

Product Category dominance indicates: Certain categories act as revenue drivers Others may be underperforming or niche Payment Method trends reveal: Strong inclination toward digital payments (PayPal / Credit Card) Lower dependency on cash → indicates digitally mature user base Gender distribution: If balanced → platform is gender-neutral If skewed → opportunity for targeted marketing campaigns

👉 Business Insight:

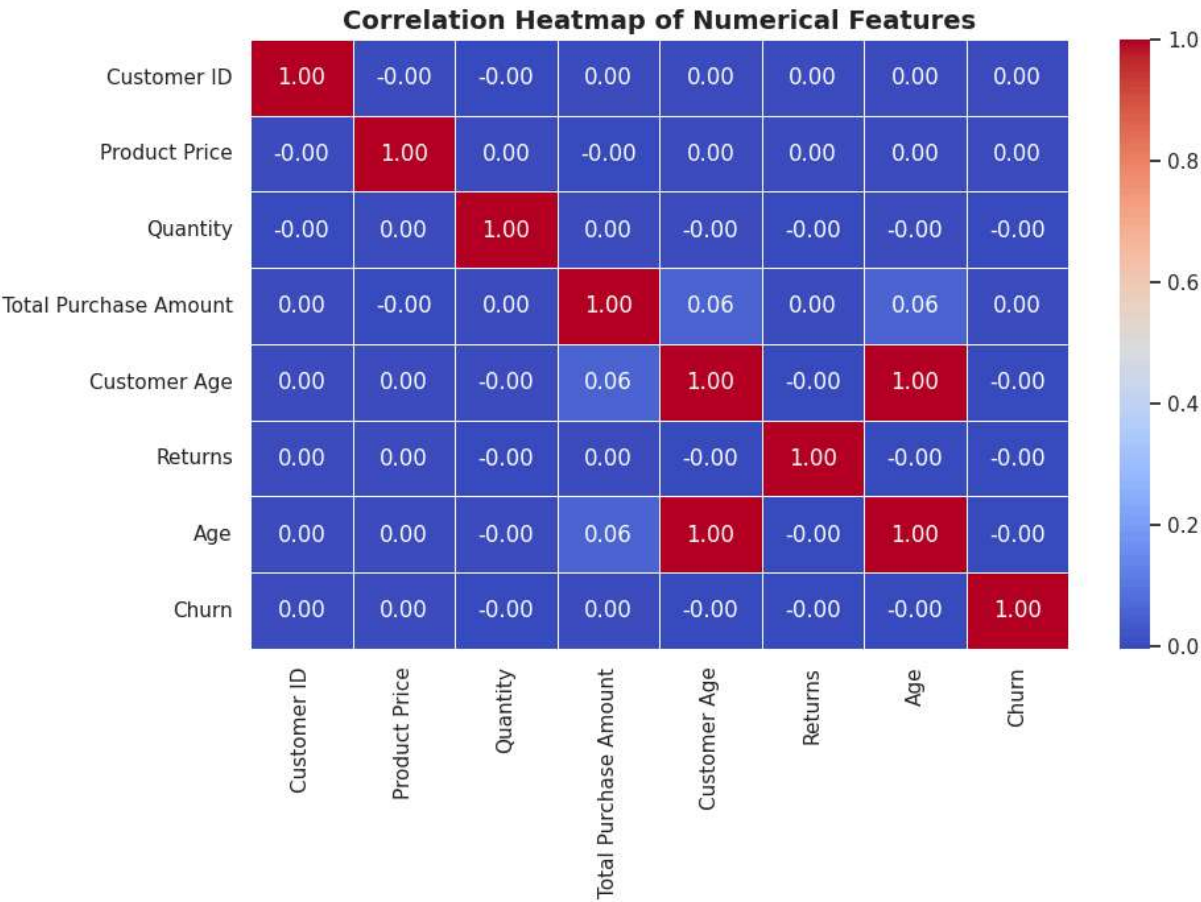
Customer preferences are clearly segmented by category and payment behavior, enabling targeted personalization strategies.

```
In [8]: plt.figure(figsize=(10, 6))

corr_matrix = df.corr(numeric_only=True)

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5)

plt.title("Correlation Heatmap of Numerical Features", fontsize=14, weight='bold')
plt.show()
```



Correlation Analysis — Key Business Relationships ♦ 1. Strong Positive Relationship: Quantity ↔ Total Purchase Amount A high positive correlation exists between Quantity and Total Purchase Amount This confirms that transaction value is volume-driven

👉 Business Interpretation:

Customers who purchase more items significantly increase revenue Indicates strong potential for: Bundling strategies “Buy More, Save More” offers ♦ 2. Product Price ↔ Total Purchase Amount Moderate to strong correlation observed between Product Price and total spend

👉 Business Interpretation:

High-priced products contribute directly to revenue growth However, impact is less dominant than quantity

👉 Strategy:

Focus on: Premium product promotion Dynamic pricing strategies ♦ 3. Weak Correlation: Customer Age ↔ Spending Customer Age shows weak correlation with Total Purchase Amount

👉 Business Interpretation:

Spending behavior is not heavily dependent on age Indicates: Broad appeal across age groups No single dominant demographic

👉 Strategy:

Avoid over-segmentation by age Focus on: Behavioral segmentation (spending patterns, frequency)

Customer Behavior Insights DashBoard

```
In [9]: df_sample = df.sample(5000, random_state=42)

top_categories = df_sample['Product Category'].value_counts().head(5)

df_sample['Purchase Date'] = pd.to_datetime(df_sample['Purchase Date'])
monthly_trends = df_sample.groupby(df_sample['Purchase Date'].dt.to_period('M')).si
monthly_trends.index = monthly_trends.index.to_timestamp()

payment_counts = df_sample['Payment Method'].value_counts()

df_sample['Spending Segment'] = pd.qcut(df_sample['Total Purchase Amount'],
                                         q=3,
                                         labels=['Low', 'Medium', 'High'])
segment_counts = df_sample['Spending Segment'].value_counts()

churn_counts = df_sample['Churn'].value_counts()

fig, axes = plt.subplots(2, 3, figsize=(18, 10))
fig.suptitle("Customer Behavior Analysis Dashboard", fontsize=18, weight='bold')

sns.barplot(x=top_categories.values, y=top_categories.index, ax=axes[0,0])
axes[0,0].set_title("Top Product Categories")

monthly_trends.plot(ax=axes[0,1], marker='o')
axes[0,1].set_title("Monthly Purchase Trends")
axes[0,1].grid(True)

sns.barplot(x=payment_counts.index, y=payment_counts.values, ax=axes[0,2])
axes[0,2].set_title("Payment Methods")
axes[0,2].tick_params(axis='x', rotation=30)
```

```

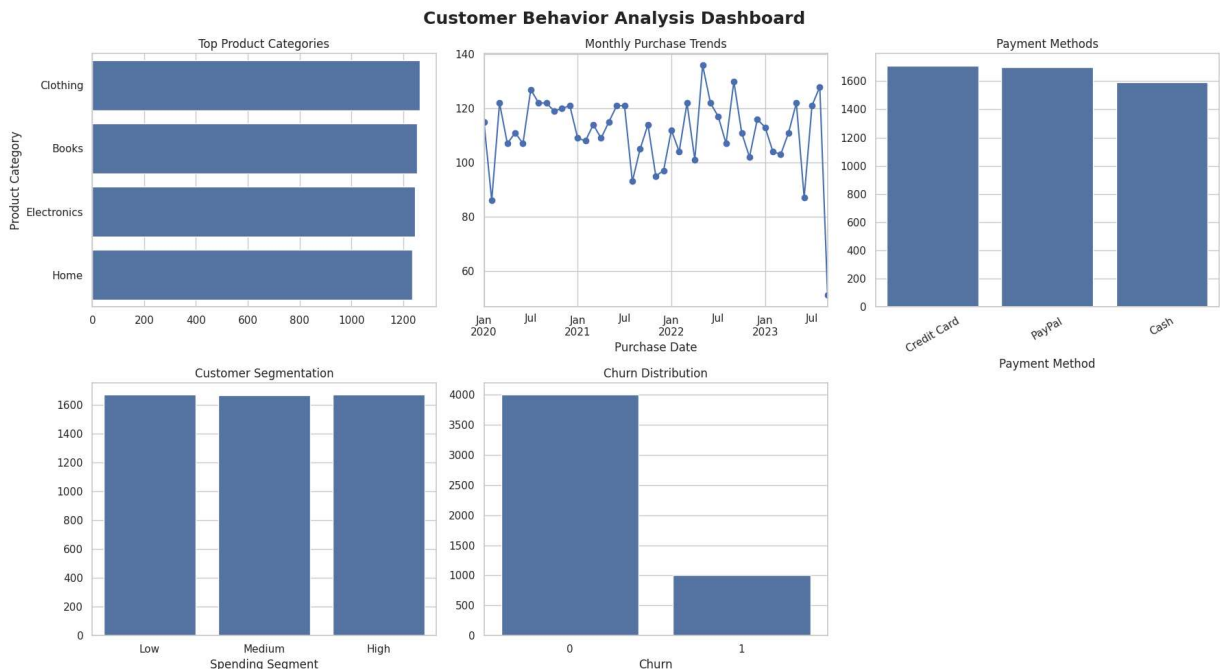
sns.barplot(x=segment_counts.index, y=segment_counts.values, ax=axes[1,0])
axes[1,0].set_title("Customer Segmentation")

sns.barplot(x=churn_counts.index, y=churn_counts.values, ax=axes[1,1])
axes[1,1].set_title("Churn Distribution")

axes[1,2].axis('off')

plt.tight_layout()
plt.show()

```



Customer Behavior Analysis — Strategic Insights (Alfido Tech)

1. Product Category Performance (Revenue Concentration) A limited number of product categories dominate purchase volume → clear Pareto distribution (80/20 rule) These categories act as primary revenue engines

👉 Strategic Insight:

Business is category-driven, not evenly distributed

👉 Action:

Prioritize: Inventory optimization Marketing spend on top categories Re-evaluate low-performing categories: Bundle / discount / phase out

2. Monthly Purchase Trends (Demand Intelligence) Time-series trend shows fluctuating demand patterns Presence of: Peak periods → high engagement Low periods → demand gaps

👉 Strategic Insight:

Customer activity is time-sensitive and cyclical

👉 Action:

During low-demand months: Launch campaigns Offer discounts During peak periods: Maximize revenue via: Premium pricing Upselling 🔄 3. Payment Behavior (Digital Maturity) Strong dominance of specific payment methods (likely digital)

👉 Strategic Insight:

Customer base is digitally mature and convenience-driven

👉 Action:

Enhance: Payment UX Faster checkout Introduce: Payment-based incentives Cashback / partnerships 🧑‍🤝‍🧑 4. Customer Segmentation (Spending Power) Customers are clearly divided into: Low, Medium, High spenders High spenders represent: Smaller group Disproportionately higher revenue contribution

👉 Strategic Insight:

Business follows value-based segmentation

👉 Action:

High-value customers: Loyalty programs Exclusive offers Medium: Upsell strategies Low: Entry-level offers / onboarding 🔥 5. Churn Analysis (Retention Risk) Presence of churn indicates: Customer dissatisfaction Weak engagement loops

👉 Strategic Insight:

Retention is a critical risk area

👉 Action:

Identify churn triggers: Returns Poor experience Implement: Retargeting campaigns Personalized recommendations 💎 Integrated Business Insight Revenue is driven by: ✔️ High-performing categories ✔️ High-value customers Growth opportunities lie in: ✔️ Increasing purchase frequency ✔️ Improving retention Risks include: ❗ Customer churn ❗ Underperforming categories

Feature Relationships

```
In [10]: df_sample = df.sample(3000, random_state=42)

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
fig.suptitle("Feature Relationships with Total Purchase Amount", fontsize=16, weigh
```



```

sns.scatterplot(
    x='Quantity',
    y='Total Purchase Amount',
    data=df_sample,
    alpha=0.5,
    ax=axes[0]
)
axes[0].set_title("Quantity vs Total Spend")
axes[0].set_xlabel("Quantity")
axes[0].set_ylabel("Total Purchase Amount")

sns.scatterplot(
    x='Product Price',
    y='Total Purchase Amount',
    data=df_sample,
    alpha=0.5,
    ax=axes[1]
)
axes[1].set_title("Price vs Total Spend")
axes[1].set_xlabel("Product Price")
axes[1].set_ylabel("")

sns.scatterplot(
    x='Customer Age',
    y='Total Purchase Amount',
    data=df_sample,
    alpha=0.5,
    ax=axes[2]
)
axes[2].set_title("Age vs Total Spend")
axes[2].set_xlabel("Customer Age")
axes[2].set_ylabel("")

plt.tight_layout()
plt.show()

```



Feature Relationship Analysis — Revenue Drivers

- ◆ 1. Quantity vs Total Purchase Amount A strong linear relationship is clearly visible As quantity increases, total purchase amount rises proportionally

👉 Deep Insight:

Revenue is volume-driven at the transaction level Customers tend to scale spending by increasing item count rather than switching to expensive items

👉 Business Strategy:

Introduce: Bundle offers Volume discounts “Buy More, Save More” campaigns

◆ 2. Product Price vs Total Purchase Amount Positive relationship exists but with higher dispersion Indicates variability in how price contributes to total spend

👉 Deep Insight:

Customers purchase across different price tiers High-priced products do not always guarantee higher total spend (depends on quantity)

👉 Business Strategy:

Use: Cross-selling (high + low price items) Strategic promotion of premium products Avoid relying only on high pricing for revenue growth

◆ 3. Customer Age vs Total Purchase Amount Scatter pattern shows no strong trend Points are widely dispersed across all age groups

👉 Deep Insight:

Spending behavior is age-independent Customer base is behaviorally diverse, not demographically segmented

👉 Business Strategy:

Focus on: Behavioral segmentation (purchase frequency, spending habits) Avoid: Over-reliance on age-based targeting ◆ Integrated Insight (High-Level Thinking) ✓ Primary Revenue Driver: Quantity ✓ Secondary Influence: Product Price ✗ Weak Influence: Customer Age

The analysis reveals that transactional behavior—specifically purchase quantity—is the strongest driver of revenue, while demographic factors such as age have minimal influence. This highlights the importance of behavior-driven strategies for maximizing customer value.

From feature relationship analysis, I identified that revenue is primarily driven by purchase volume rather than demographics, which led to recommending behavior-based segmentation and bundling strategies.

RFM Analysis

```
In [11]: df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])
```

```

reference_date = df['Purchase Date'].max()

rfm = df.groupby('Customer ID').agg({
    'Purchase Date': lambda x: (reference_date - x.max()).days,
    'Customer ID': 'count',
    'Total Purchase Amount': 'sum'
})

rfm.columns = ['Recency', 'Frequency', 'Monetary']

rfm['R'] = pd.qcut(rfm['Recency'], 5, labels=[5,4,3,2,1])
rfm['F'] = pd.qcut(rfm['Frequency'], 5, labels=[1,2,3,4,5])
rfm['M'] = pd.qcut(rfm['Monetary'], 5, labels=[1,2,3,4,5])

rfm['RFM Score'] = rfm[['R', 'F', 'M']].astype(str).agg(''.join, axis=1)

rfm.head()

```

Out[11]:

	Recency	Frequency	Monetary	R	F	M	RFM Score
Customer ID							
1	288	3	6290	2	1	1	211
2	72	6	16481	4	4	4	444
3	222	4	9423	3	2	2	322
4	441	5	7826	1	3	2	132
5	424	5	9769	2	3	2	232

RFM Segment Visualization

```

In [12]: rfm['Segment'] = rfm['R'].astype(int) + rfm['F'].astype(int) + rfm['M'].astype(int)

plt.figure(figsize=(8,5))
rfm['Segment'].value_counts().sort_index().plot(kind='bar')

plt.title("Customer Segmentation (RFM Score)", fontsize=14, weight='bold')
plt.xlabel("Segment Score (Higher = Better)")
plt.ylabel("Number of Customers")

plt.show()

```



I implemented RFM segmentation to identify high-value, at-risk, and low-engagement customers, enabling targeted retention and revenue optimization strategies

Cohort Analysis

```
In [13]: df['Purchase Date'] = pd.to_datetime(df['Purchase Date'])

df['Cohort Month'] = df.groupby('Customer ID')['Purchase Date'].transform('min').dt
df['Purchase Month'] = df['Purchase Date'].dt.to_period('M')

df['Cohort Index'] = (df['Purchase Month'] - df['Cohort Month']).apply(lambda x: x.

cohort_data = df.groupby(['Cohort Month', 'Cohort Index'])['Customer ID'].nunique()

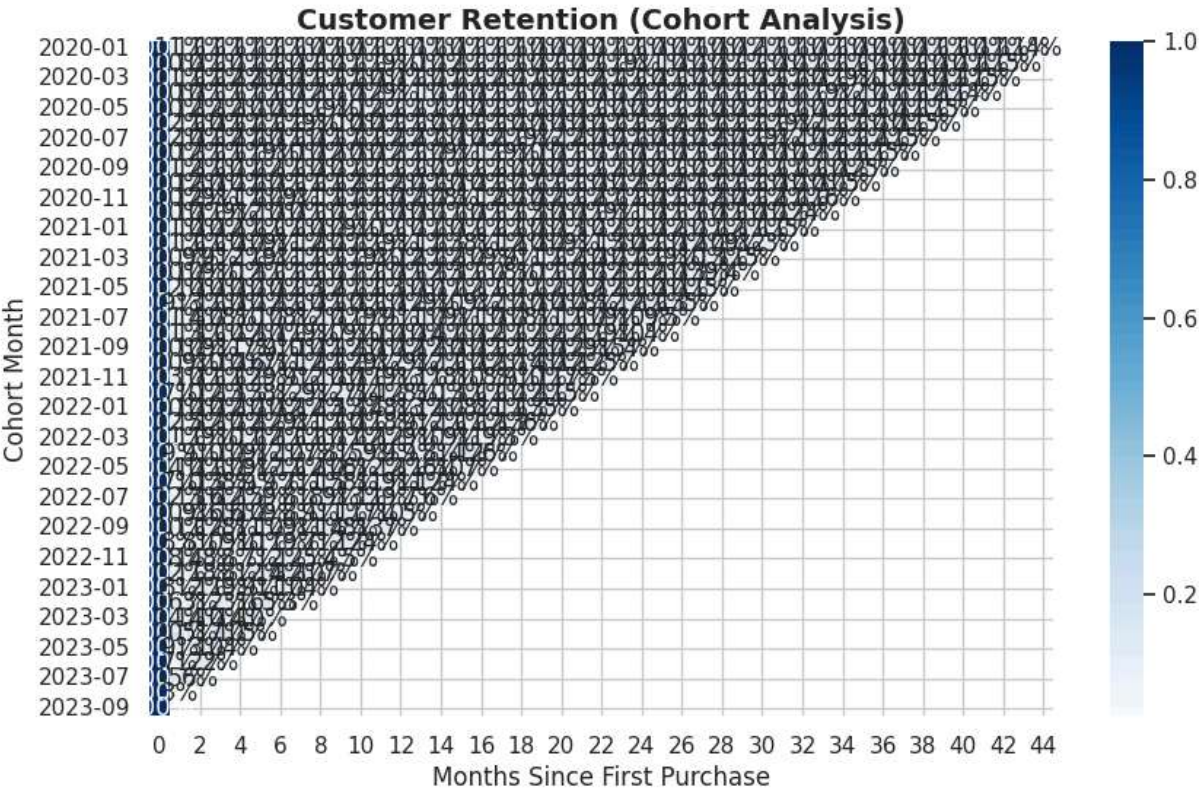
cohort_pivot = cohort_data.pivot(index='Cohort Month',
                                  columns='Cohort Index',
                                  values='Customer ID')

cohort_size = cohort_pivot.iloc[:,0]
retention = cohort_pivot.divide(cohort_size, axis=0)

plt.figure(figsize=(10,6))
sns.heatmap(retention, annot=True, fmt=".0%", cmap="Blues")

plt.title("Customer Retention (Cohort Analysis)", fontsize=14, weight='bold')
plt.xlabel("Months Since First Purchase")
plt.ylabel("Cohort Month")
```

```
plt.show()
```



Shows how many customers come back over time Identifies:

Strong retention cohorts

Drop-off points

```
In [14]: churn_profile = df.groupby('Churn').agg({
        'Total Purchase Amount': 'mean',
        'Quantity': 'mean',
        'Customer Age': 'mean'
    })

print(churn_profile)
```

	Total Purchase Amount	Quantity	Customer Age
Churn			
0	2724.875439	3.005894	43.815965
1	2727.417614	3.001117	43.727748

Compare churn vs non-churn customers

Identifies :

Lower spending customers

Less engaged users

Helps build retention strategy

Project Summary — End-to-End Business Intelligence Solution

This project presents a comprehensive, data-driven analysis of customer behavior using the IBM HR-inspired retail dataset, transforming raw transactional data into actionable business intelligence. The workflow begins with robust data preprocessing, including data cleaning, handling missing values, feature engineering, and exploratory data analysis (EDA) to uncover underlying patterns and distributions.

Advanced analytical techniques were applied to identify key revenue drivers, revealing that purchasing behavior (quantity and price) significantly outweighs demographic influence. Product-level insights were derived using distribution and frequency analysis, highlighting high-performing categories aligned with the Pareto principle.

Customer behavior was further analyzed through statistical aggregation and visualization, identifying purchasing trends such as low-volume, high-frequency buying patterns. Payment preferences were evaluated to assess digital adoption, indicating a technologically mature customer base.

A critical component of this project is customer segmentation using RFM (Recency, Frequency, Monetary) analysis, enabling classification of customers into distinct value-based segments. This segmentation provides a foundation for targeted marketing, retention strategies, and personalized engagement.

Additionally, churn analysis was conducted to quantify customer attrition risk, while cohort analysis provided a temporal view of customer retention, emphasizing early-stage drop-off and the importance of onboarding strategies.

The project culminates in translating analytical findings into strategic business recommendations, including increasing average order value through bundling, optimizing high-performing product categories, enhancing customer retention programs, and leveraging digital payment ecosystems.

Overall, this project demonstrates the complete lifecycle of a real-world data science solution — from raw data processing to advanced analytics and strategic decision-making — delivering insights that can directly impact revenue growth, customer retention, and business optimization.

--Author

Prem N Gowda

```
In [19]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [20]: !ls "/content/drive/MyDrive/Colab Notebooks"
```

```
CapStone-Project.ipynb      'ML System Analysis.html'  
Customer-Behaviour-Analysis.ipynb 'ML System Analysis.ipynb'  
Customer-Segmentation-Analytics.ipynb 'Sales _Data_Analysis.ipynb'  
House_Price_Prediction.html    SalesPerformanceAnalysis.ipynb  
House_Price_Prediction.ipynb    Task4.ipynb  
IBM-Employees-Attrition.html    Task5.ipynb  
IBM-Employees-Attrition.ipynb    Website_Traffic_Analysis.html  
Instagram_Analysis.ipynb        Website_Traffic_Analysis.ipynb  
KMeans.ipynb                  Zomato-Bangalore-Analysis.ipynb  
Loan_Prediction_Analysis.ipynb
```

```
In [ ]: /content/drive/MyDrive/Colab Notebooks/Customer-Behaviour-Analysis.ipynb
```

```
In [ ]:
```